Akanksha Sachan

Contact

Joint Carnegie Mellon University-University of Pittsburgh PhD Program | aks203@pitt.edu
Aging Institute & Dept. of Immunology, School of Medicine, UPitt | LinkedIn
Machine learning for Genomics & Epigenetics | Twitter | Github | Google Scholar

Education

University of Pittsburgh — *'24 – Present*
PhD Candidate in Computational Biology
Advised by Dr. Aditi Gurkar & Dr. Jishnu Das

Carnegie Mellon University — *'22 – '24*
PhD Student in Computational Biology
Comp-Bio Department, School of Computer Science | GPA: 3.58/4

Indian Institute of Technology (IIT), Bombay — *'18 – '22*
BTech. in Chemical Engineering | GPA: 8.06/10

Awards and Honors

- **American Federation for Aging Research** (AFAR) Graduate Scholarship — '25
- **Best Poster** Award at the Aging Institute Research Day, UPitt — '24
- **1st Place, Student A100 track** NeurIPS LLM Efficiency Challenge | team of 5 — '23
- **$500 Track Prize** (Healthcare Accessibility for Underrepresented Communities), Pitt Challenge Hackathon (6th annual) | team of 3 — '22
- Recipient of **Undergraduate Research Award** (URA-01) by IIT Bombay — '22
- Recipient of Merit-cum-Means (MCM) Scholarship by IIT Bombay for two semesters for financial hardship — '21
- Awarded a full scholarship ($5,000) for iSURE undergraduate research program by the University of Notre Dame — '20
- **Top 0.2%** (Rank 3188 out of **1.1M+** candidates) in JEE (Main) — '18
- **Top 1.5%** (Rank 2517 out of **160k+** candidates) in JEE (Advanced) — '18

Invited Talks

"FIREFate: Functional and Interpretable Regulatory Encodings of cell Fate" | *paper-in-prep*

- Keystone Symposia: Artificial Intelligence in Molecular Biology — *Sep '25*
- International Society for Computational Biology (ISCB)/(ISMB) — *Jul '25, '26*
- American Association of Immunologists (AAI) Annual Meeting 2026 — *Apr '26*
- Center of Lung Aging and Regeneration, School of Medicine, UPitt — *Nov '25*
- 30th Annual School of Medicine Graduate Student Research Symposium, UPitt — *Oct '25*
- Joint CMU-UPitt Ph.D. Program in Computational Biology, Orientation Retreat — *Aug '25*

"Metabolic rewiring of skeletal muscle driven by loss of DNA-repair" | *paper-in-prep*

- Center for Systems Immunology Annual Retreat, UPitt — *Nov '25*
- 23rd Annual Department of Immunology Scientific Retreat, UPitt — *Sep '25*
- Aging Institute Research Day, UPitt — *Nov '24*

"Micro-MPA digital twin, with sclerostin-dependent RANKL production reproduces pivotal dual-action effect of romosozumab found in clinical trials"

- 50th European Calcified Tissue Society (ECTS) Congress — *Apr '23*

Athletic Accolades

Bronze Medal | 35th Intercollegiate (inter-IIT) Aquatics Meet | 3rd in Medley-Relay *'19*
3rd Position | Comp-Bio Department, Pretty Good Race (5km) for SCS, CMU *'23*
Swam 12-hours | Swimathon organized by the IIT Bombay aquatics committee *'19*
Ran 26.2 miles | Dick's Sporting Goods Pittsburgh Marathon | 5 hours 15 mins *'24*
Ran 13.1 miles | Dick's Sporting Goods Pittsburgh Half Marathon | 2 hours 30 mins *'23*

Leadership & Societal Contributions

Vice President *Aug '25 – Present*
Allegheny Science Policy and Governance (ASPG)
Leading a recognized chapter of the **National Science Policy Network**
Bridged scientists and PA legislators on policies about **AI/data-centers and Aging** at the State Capitol, Harrisburg, PA

Technights Coordinator *Aug '23 – Aug '25*
Carnegie Mellon University, Robotics Institute
Organized and led a **computer-science outreach** program teaching middle-school girls sessions on foundation models, and deep learning.

NIH Consortium Hackathon, Co-organizer & Team Lead *Mar '24*
4D Nucleome, Seattle, Washington
Led a team of 5 to build integrative pipelines for single-cell HiC analysis from cell-atlas-scale datasets

Reviewer
ICML Foundation Model for Life Sciences Workshop 2026, PNAS Journal (assisted PI), Aging Cell Journal (assisted PI)

Teaching Assistant *Fall '23 – Spring '24*
02-760: Lab Methods for Computational Biologists, SCS, CMU
Taught a lecture on **genomics technologies** to a class of ∼20 students, revamping the RNA sequencing module of the course.

Treasurer • Senator *Aug '23 – Aug '25*
Graduate Student Association (Joint CMU-UPitt Comp-Bio PhD Program)

Editorial Board Member • Design Convenor *'19 – '21*
Insight, the student newsletter of IIT Bombay

Convenor, Girls Swim Team *'19 – '20*
Aquatics, Indian Institute of Technology Bombay

Consortiums & Workshops

NIH Consortium Annual Meetings *'23, '25*
IGVF, SenNet:Cellular Senescence Network, 4D Nucleome

Grace Hopper Celebration *Oct '24*
AnitaB.org, Philadelphia

Computational Genomics Summer Institute *Jul '23*
University of California, Los Angeles

Grad Cohort for Women *April '23*
Computing Research Association (CRA-WP), San Francisco

Research Experience Prior to Thesis

Mapping the epigenetic landscape of single-cells using 3-D chromatin architecture — ’22 – ’24
Supervisor – Prof. Jian Ma, Computational Biology Department, SCS, CMU
Developed a framework to identify cis-regulatory 3-D genomic hubs controlling target-gene expression and key transcription factors in healthy and cancerous mouse-brain single-cell, multi-omic profiling.

Translational Mechanobiology Lab, MBI, National University of Singapore — ’22
Summer Internship | Supervisor – Prof. Hanry Yu, Dept. of Physiology, YLL School of Medicine
Built a Generative-Adversarial-Network (GAN) pipeline for colorectal-cancer imaging to support clinical deep-learning deployment

Lab for Bone Biomechanics, Institute of Biomechanics, ETH Zurich — ’21
Summer Internship | Supervisor – Prof. Ralph Muller, Dept. of Health Sci&Tech
Added a Type-2 Diabetes pathway to an agent-based osteoporosis model and ran large-scale simulations on the CSCS Swiss National Supercomputing Center

Multicellular Systems Engineering Lab, University of Notre Dame — ’20
Summer Internship | Supervisor – Prof. Jeremiah Zartman, Chemical & Biomolecular Engineering
Built a morphometric feature-extraction pipeline for *Drosophila* wing imaginal discs (OpenCV, Fiji) to study organism-wide cell-signaling

Key Coursework

Machine Learning & Statistics: Intermediary Statistics (CMU: 36-705), Probabilistic Graphical Models (CMU: 10-708), Introduction to Machine Learning (CMU: 10-701)

Predicting genomic-hub mediated TF-perturbation response via geometric deep learning — ’23
10-708: Probabilistic Graphical Models
Augmented a GNN for Perturb-seq response prediction with a Hi-C contact graph (Cancer cell-line), adding 3D-genome topology to the co-expression and Gene Ontology graph features.

Single-cell DNA methylation prediction using single-cell HiC data and Graph NNs — ’23
02-710: Computational Genomics
Developed a **GNN** to predict methylation from HiC data, connecting structure to phenotype; clustered **single-nucleus DNA methylation** data with K-means and t-SNE.

Compute Allocation Grants

ACCESS (NSF Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support) allocation on the NCSA Delta GPU supercomputer — ’26